

Layer 1: geometry and rollout stability

clean train / clean eval, 60 s horizon

Full-State BB

no pose geometry

5/5 diverged

no finite median



retain manifold structure

SE(3) Momentum BB

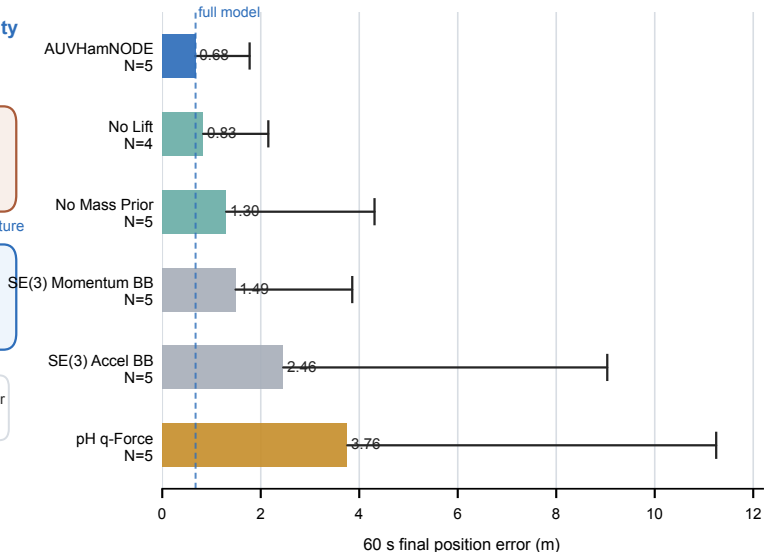
SE(3)+mass

1.49 m

median; P95 3.86 m; N=5

Evidence claim: geometry is a key stability factor within this model family.

Layer 2: energy core and precision



Bars show seed-mean terminal medians; whiskers show seed-mean P95. No Lift excludes seed 43 by the training-anomaly rule.